

臺北市立大學 資訊科學系 <u>114</u> 學年度 第 <u>2</u> 學期 <u>資料結構</u> 第二次期中考試卷			
院系	_____學院 _____系 _____年 _____班		
學號		姓名	

注意事項：

1. 考試時間：15:30~19:30
2. 不需要按照題號順序作答，但請在答案卷上將題號標示清楚
3. 請盡量作答，寫出計算過程
4. 字跡請盡量工整
5. 若空間不夠書寫請舉手，助教會提供新的答案卷。

1. (25%) Please prove or disprove the following each statement.
 - (1) (10%) Given a one dimensional array, we can construct a max-heap.
 - (2) (10%) Let T_1 be a complete binary tree with n nodes. Then the height of T_1 is $\lfloor \log_2 n \rfloor + 1$.
 - (3) (5%) Given preorder traversal and postorder traversal of tree T_2 , we reconstruct the unique original tree.
2. (15%) Given the following key values, please construct the height-balance binary tree. (Note: Draw the AVL tree following each insertion and state the rotation type (if any) for each insert.)
12, 1, 5, 3, 7, 6, 4, 16, 13
3. (5%) Given the last four digits of your student ID number, determining the m according the following equation:

$$m = ((\text{the last four digits}) \bmod 199) + 41$$

NOTE: The value m will be used in the following problems.

4. (30%) Suppose that we have the following key values: 7, 16, 49, 82, m , 5, 31, 6, 2, 46. Please construct the following different trees. **Write out the result after each value is inserted into the trees.**
 - (1) (10%) binary search tree
 - (2) (10%) min-heap
 - (3) (10%) complete binary tree

You MUST write down your answers step by step for Problem 5 and Problem 6, otherwise, you cannot obtain the full grade points.

5. **(30%)** Please illustrate the operations of **(1)(10%) HEAP SORT**, **(2)(10%) BINARY TREE SORT**, **(3)(10%) RADIX SORT**, respectively, on the array $A = \{152, 208, 306, 93, m, 859, 984, 55, 9, 271, 33\}$.

NOTE: There are two methods for RADIX SORT.

6. **(15%)** Please illustrate the operations of **(1)(5%) LINEAR SEARCH**, **(2)(10%) BINARY SEARCH**, respectively, on the array $A = \{179, 208, 306, 93, m, 859, 984, 55, 9, 271, 33\}$. The searching key is m and 550.

You MUST write down your answers step by step for Problem 5 and Problem 6, otherwise, you cannot obtain the full grade points.